Class Work- 01/04/2026

Part A: Understanding the Dataset (Exploration)

1. Load the CSV file into pandas and display the first 10 rows.

Students must demonstrate:

- importing pandas
- reading CSV
- previewing data

2. Determine:

- number of rows
- number of columns
- variable data types

Then explain:

Which variables are numerical and which are categorical?

3. Check whether missing values exist.

Then answer:

If missing values existed, which method would you choose to handle them and why?

Part B: Descriptive Analysis

4. Calculate:

- average age
- average BMI
- average glucose level

Then explain:

Which variable shows the highest spread?

5. Count how many patients belong to each diagnosis category.

Then answer:

Which diagnosis appears most frequently?

Part C: Visualization

6. Plot a histogram of age.

Then explain:

Is the age distribution balanced or skewed?

7. Plot blood pressure against age.

Then answer:

Does blood pressure appear to increase with age?

8. Compare BMI across diagnosis groups using a boxplot.

Then interpret:

Which diagnosis group appears to have higher BMI?

Part D: Pattern Discovery

9. Compute correlation between:

- Blood Pressure
- Cholesterol
- Glucose
- BMI

Then explain:

Which two variables are most strongly related?

10. Create a new variable:

Risk_Level

Rules:

- High if BMI > 30 and Glucose > 150
- Medium if one condition is true
- Low otherwise

Then count how many patients fall into each category.

Part E: Machine Learning Introduction

11. Use one simple classification model to predict Diagnosis.

Suggested:

- Decision Tree
  or
- Logistic Regression

Students must:

- split training/testing data
- fit model
- report accuracy

12. Explain:

Why accuracy alone may not fully describe model quality?

Part F: Critical Thinking

13. Ask GenAI to suggest one machine learning model for this dataset.

Then explain:

- why the suggestion is useful
- what the AI ignored
- what you would improve

Part G: Local Reflection

14. If this dataset came from a district hospital near your home, what extra variable would improve prediction quality?